PKM Kummer tower at P-224's 2-adic shape

The PKM Kummer tower at P-224's 2-adic shape, priced against matched rho

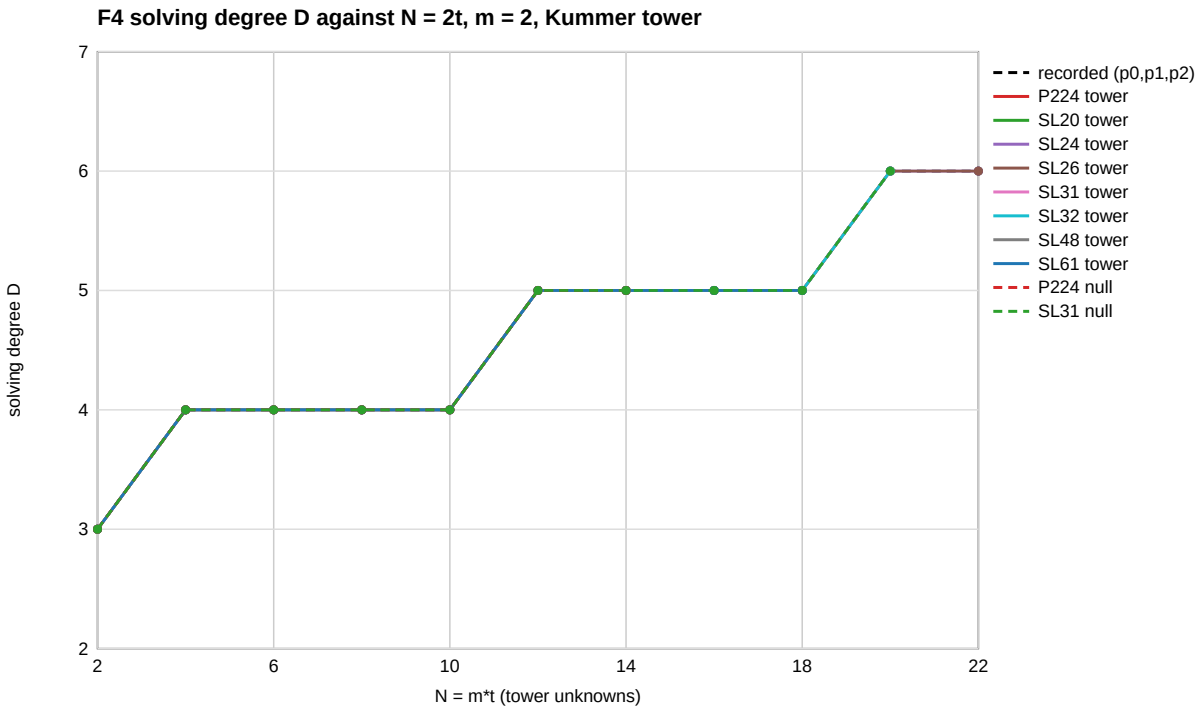
2026-10-22. Branch `claude/p224-pkm-kummer-10gpep`. Protocol: PROTOCOL.md, frozen in commit `375893d02` before any measured run; two dated amendments (arms, before any ecbench session; R18 and R18b, before R18b) are in the same file. Lead: claim 2d of research/p224_hypothesis_ranking_20261008/. Background: RESEARCH_PKM_TOWER_ORACLE.md §§10-17.

Every number is labelled **measured** (this directory's runs), **recorded** (an earlier committed row), **derived** (a calculation shown) or **predicted** (stated in the protocol before the runs).

Verdict

Inconclusive on the solver axis; clean negative on price; no improvement over the index-calculus baseline. Not an advance, not engineering, not relabelling. One accounting artefact is named in §3.4 so it is not mistaken for a gain.

- **Measured, solver axis (stage diagnostic, no S).** At eight P-224-shaped Solinas primes `p = 2^n - 2^k + 1` (20 to 61 bits) and at the NIST P-224 prime itself, the F4 solving degree `D` of the Kummer tower equals the recorded `D` in all 89 valued tower cells (`m = 2, 3, 4`), the width and step count of every system equal the recorded ones, and the shape-matched null gives the same `D` in all 65 paired cells. `k/n` ran from 0.42 to 0.70 and the odd part of `p - 1` from 6 to 128 bits; the prior pilot primes had `k/n` 0.87-0.94 and a 2-4 bit odd part. So the P-224 *proportions* of 2-adic structure change nothing F4 can see. H_same holds (predicted).
- **Measured, the one cell past the recorded frontier did not finish.** `m = 2, N = 24` (`t = 12`) was killed by the kernel's out-of-memory killer at 6.0 GB (R18, below), so it has no `D`. A capped re-run (R18b) gives only `D >= 6`. The `m = 2` line over `N = 2...22` reads **inconclusive** under rule A1 (plateau length 2, upper-half slope 0.114). Question Q2 stays open; it needs a host with more than this one's memory (its `B'` block passes 7.9e8 entries at step 26 of an unknown number).
- **Measured, price (ecbench, full cost, matched rho).** On four Solinas curves of 20 to 28 bits (and four Proth controls) the Kummer-base index calculus costs **18 to 78 times** rho with the negation map (`S` ratio, `w = 0`), **19.9 to 134 at `w = 1`**; rho is below every index-calculus arm on every curve at every weight of the band. Every one of the 1,152 runs (warm-ups included) verified and 864 replays reproduced.
- **Derived, the F4 method priced end to end at the one cell that matches a curve** (20 bits, `m = 2, N = 20`): at least **2.4 × 10^9** in `S`, about **1.9 × 10^9 times** matched rho at `w = 0` and **1.0 x 10^9** at `w = 1` (generous bound: 6 × 10^10 `S` with one field inversion costed as one multiplication). Only one F4 row exists, so there is no scaling fit for it.
- **Distance to rho (success criterion).** Best baseline-family arm: 18.4 times rho (20-bit Solinas, `ic-mitm`) and 75.9 times at 28 bits. At equal `t` the Solinas curves cost 2% to 32% more in absolute GAE than the prior ladder's Proth curves (§3.4), so no row moved toward rho.



Measured rows of runs/*.jsonl (pkm_p224_kummer_shape_20261022); recorded = PKM note 11-12, 16. Stage diagnostic, no S.

1. What ran

run	what	status (measured)
R0	positive control: isogeny kind, `p0`, `m = 2`, `N = 4-10`	**passed**: `D = 5` on both targets at every `N`
R1-R3	Kummer `m = 2`, `N = 2-20`, tower and null, SL26, SL31, SL32	complete, 41 rows each
R4	SL20, `m = 2`, `N = 16-20` (the 20-bit priced curve's prime)	complete
R5-R8	`m = 3` (`N = 6, 9, 12`) and `m = 4` (`N = 8, 12`), tower and chain null	complete
R9-R12	SL48 (Fp64) and SL61 (Fp64), `m = 2` to `N = 18`, `m = 3` to `N = 12`, `m = 4` `N = 8`	complete
R13-R15	P-224 (Fp224): `m = 2` to `N = 16` and `N = 18`, `m = 3` to `N = 12`, `m = 4` `N = 8`; wide builder `--verify`	complete, 23/23 verified in `BigUint`
R16	SL24, `m = 2`, `N = 20` and 22, one target each; null at `N = 22` (R16n)	complete
R17	SL31, `m = 3`, `N = 15`, one target	complete, `D = 7`
R19	SL26, `m = 2`, `N = 22`, one target	complete
R18	SL24, `m = 2`, **`N = 24`**, 2 threads, cap 1.8e9 entries	**killed (OOM), no row, no D** (§4)
R18b	the same system, `--max-nnz 6e8 --trace` (Amendment 2)	stopped at step 26, `D >= 6`
s20-s28	ecbench sessions, Solinas and Proth curve per size	4 sessions, all audits OK

Where `N = 22` was measured (SL24 and SL26, one target each), `D = 6`, the recorded value. The recorded `m = 2` frontier was `N = 22`; this round did not pass it.

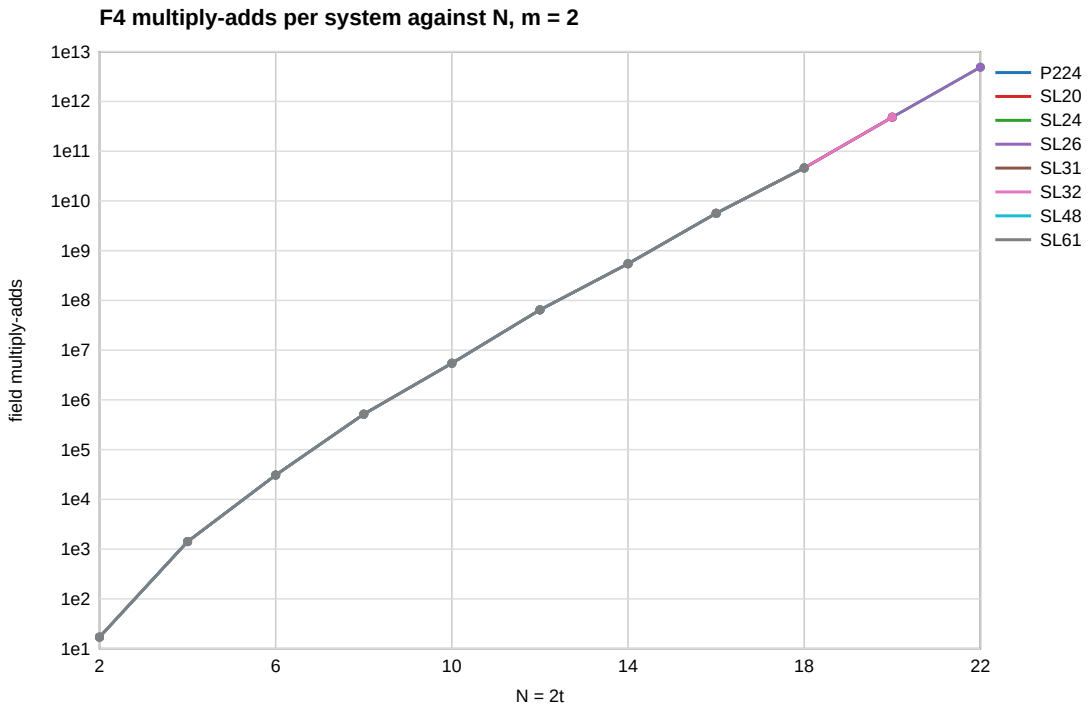
2. Solver axis (measured; stage diagnostic)

The tower is the Kummer tower `V = mu_{2^t}` (`g = 1`), `f4_tower_generic` at `Fp32`, `Fp64` and `Fp224`, two random targets per cell unless noted, seed 1347112276. Cell value: `D` when every target finished with the same `D`.

m	N	recorded D	SL20	SL24	SL26	SL31	SL32	SL48	SL61	P224	null D (same cell)
2	2	3	·	·	3	3	3	3	3	3	3 at every prime run
2	4	4	·	·	4	4	4	4	4	4	4
2	6	4	·	·	4	4	4	4	4	4	4
2	8	4	·	·	4	4	4	4	4	4	4
2	10	4	·	·	4	4	4	4	4	4	4
2	12	5	·	·	5	5	5	5	5	5	5
2	14	5	·	·	5	5	5	5	5	5	5
2	16	5	5	·	5	5	5	5	5	5	5
2	18	5	5	·	5	5	5	5	5	5	5
2	20	6	6	6	6	6	6	·	·	·	6 (SL26, SL31, SL32)
2	22	6	·	6	6	·	·	·	·	·	6 (SL24)
2	**24**	no value	·	**stopped, D ≥ 6**	·	·	·	·	·	·	·
3	6	5	·	·	5	5	5	5	5	5	5
3	9	6	·	·	6	6	6	6	6	6	6
3	12	6	·	·	6	6	6	6	6	6	6
3	15	7	·	·	·	7	·	·	·	·	not run
4	8	6	·	·	6	6	·	6	6	6	6
4	12	7	·	·	7	7	·	·	·	·	7

· = not run. Full per-cell table (targets, width, steps, multiply-adds, time): `tables/f4_cells.md` and `tables/f4_cells.tsv`; decision lines: `tables/f4_summary.txt`.

- **Q1, H_same: held.** 89 of 89 valued tower cells equal the recorded `D`; 0 differ. **Q3:** tower `D` equals null `D` in 65 of 65 paired cells (H1b does not hold: the tower is not below the null). **Q4:** at every `(m, N)` run at two or more primes (16 cells) the width and step count are equal, and the multiply-add count of target 0 agrees to within 3 × 10^-6 across the primes (`m = 2, N = 18`: 45,950,099,888 to 45,950,147,498 at seven primes from 20 to 224 bits; one thread, so the chunking is the same).
- **Positive control (R0): passed**, so the instrument can see a tower-family difference (isogeny `D = 5` against Kummer 4 at the same `N`).
- **Independent check.** `pkm_tower_check verify` recounted every finished `u64`-builder tower row by exhaustive search over `V^m`: 104 + 52 rows (R0 included), all agree (`verify_u64.txt`, `verify_u64_wide48_61.txt`). The P-224 rows (above `2^64`) were checked by the pilot's own `--verify` in `BigUint` arithmetic: 23 of 23 tower rows ok (the independent checker's integer type does not hold `p`, so it does not read those rows).
- **Q2 (frontier).** Pooling the `m = 2` tower rows over primes (a derived input file, `analysis/pooled_m2_kummer_tower.jsonl`, `p` relabelled to 1, `target_index` offset per prime, nothing else changed) and running `pkm_tower_check analyze` gives for `N = 2...22`: *A1 inconclusive (upper-half slope 0.114 over N = 12...22, final plateau L = 2)* (`analysis/pooled_m2_analyze.txt`). R18b's trace, `runs/R18b-SL24-m2-N24-sizing.log`: 25 steps ran, step 23 was at degree 6 and added 2,042 elements, so **`D >= 6`**. Step 26 (degree 6) stopped at `B'` = 7.86e8 entries, above the 6e8 cap. The recorded `N = 22` system had 28 steps in all. Nothing here reads 7.



Measured; mean over the cell's finished targets. Counts depend on the thread count (chunking).

Scope. `t <= 12` at `N <= 24`. P-224's `p - 1` has `v2 = 96`, so a tower could in principle be `t = 96`. PKM needs `|V|^m` of the order of `r`, that is `N = m t` near `log2 r = 224` at any `m`. The measured `N <= 24` is about 11% of that, and every deeper cell is extrapolation (not made here).

3. Price (measured arms; derived F4 arm)

3.1 Boundary, reference, unit

`S = total GAE / sqrt(r)`; the floor is `sqrt(pi/4) = 0.886`; the reference is `rho.negation` in the same session and binary, matched to each IC arm's workload and round. All 8 workloads per curve are one-target; 3 measured rounds; every IC arm charges every group operation.

3.2 Curves

Curves are named here by their field and parameters, not by an ICV1 slug (see the note below the table). `y^2 = x^3 + ax + b`, `#E = h r`, `r` prime.

bits	Solinas curve `(p, a, b)`	k	h	log2 r	t	Proth control `(p, a, b)` (prior ladder)	log2 r
20	(1032193 = 2^20-2^14+1, 115752, 453155)	14	1	19.98	8	(946177, 304973, 530845)	18.85
24	(16760833 = 2^24-2^14+1, 10096653, 5158538)	14	4	22.00	9	(10940417, 10128206, 867458)	23.38
26	(67104769 = 2^26-2^12+1, 36772902, 59108565)	12	2	25.00	10	(40857601, 8702713, 9617118)	23.28
28	(268369921 = 2^28-2^16+1, 147431093, 254493867)	16	1	28.00	11	(231657473, 95419558, 159693275)	26.20

Not registered. AGENTS.md §11 asks for registration before a curve's slug is cited. The four Solinas curves are not in `docs/curves/registry.json`: a registry change moves the registry digest that the cover, trait and transfer certificates bind, so it needs those regenerated

(curve_cover_check, curve_traits, jacobian_transfers, the alias map), which is a separate, large change. So no slug is written in Markdown here (the slugs are in the sealed sessions' plan.json, spec.json and records, and ecbench plan says registered=false for them). Registering them is a follow-up. The four Proth controls are registered (the prior ladder).

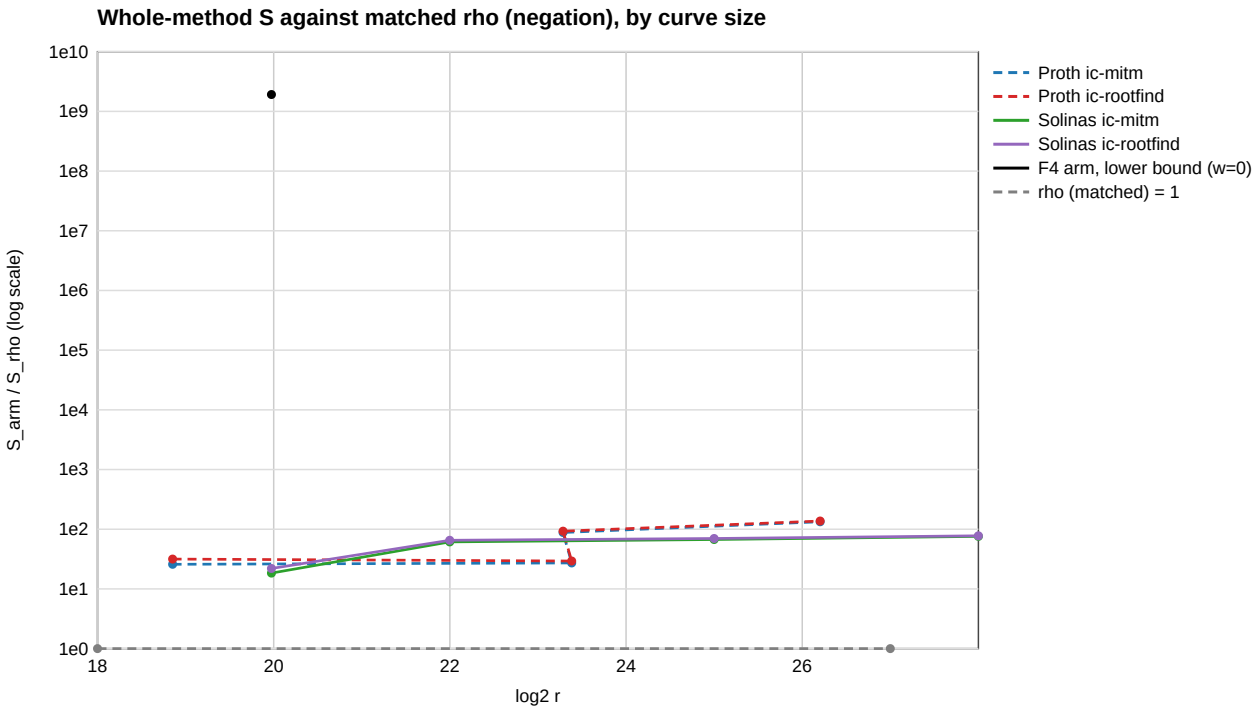
3.3 Table: S per arm, ratio to matched rho, with the §5a band

S at w = 0 / 0.1 / 1 GAE per unpriced native operation (the harness's ecbench band, reproduced by tables/ecbench_arms.md); ratio = S_arm / S_rho at the same w. The F4 arm is **derived**, one row, a lower bound (§3.5). D is the F4 solving degree of the cell the arm's size corresponds to (stage diagnostic).

curve	arm	D (stage)	S w=0 / 0.1 / 1	ratio to rho w=0 / 0.1 / 1	unpriced counters (mean)
20-bit Solinas	rho-neg		1.250 / 1.358 / 2.337	1.0	canonicalisations 1,104
20-bit Solinas	ic-mitm		22.94 / 25.30 / 46.53	18.4 / 18.6 / 19.9	lookups 6,092; row ops 17,357; 3 more
20-bit Solinas	ic-rootfind		27.54 / 29.96 / 51.69	22.0 / 22.1 / 22.1	lookups 6,167; row ops 17,845; 3 more
20-bit Solinas	**F4 arm (derived), m=2, N=20**	6	2.40e9 / 2.40e9 / 2.40e9	1.9e9 / 1.8e9 / 1.0e9	F4 pairs, reducers, rows (§3.5)
24-bit Solinas	rho-neg		1.057 / 1.155 / 2.030	1.0	
24-bit Solinas	ic-mitm		64.9 / 73.8 / 153.7	61.4 / 63.9 / 75.7	
24-bit Solinas	ic-rootfind		68.9 / 77.7 / 157.7	65.1 / 67.3 / 77.7	
24-bit Solinas	F4 arm	not priced: N = 24 has no D			
26-bit Solinas	rho-neg		0.913 / 0.998 / 1.762	1.0	
26-bit Solinas	ic-mitm		61.2 / 74.6 / 194.8	67.1 / 74.7 / 110.6	
26-bit Solinas	ic-rootfind		63.8 / 77.1 / 196.4	69.9 / 77.3 / 111.4	
28-bit Solinas	rho-neg		0.923 / 1.013 / 1.824	1.0	
28-bit Solinas	ic-mitm		70.1 / 87.5 / 244.0	75.9 / 86.3 / 133.8	
28-bit Solinas	ic-rootfind		71.8 / 89.0 / 243.3	77.8 / 87.8 / 133.4	
20-28-bit Proth controls	rho-neg, ic-mitm, ic-rootfind		see tables/ecbench_arms.md	25.9-133.7 (w=0)	

Paired ecbench compare intervals (95% cluster bootstrap over 8 workloads, per curve, w = 0), ic-mitm / rho-neg: 18.4 [14.7, 23.9] (20-bit Solinas), 61.4 [49.9, 78.0] (24), 67.0 [53.1, 86.4] (26), 75.9 [60.5, 97.4] (28); ic-rootfind / rho-neg: 22.0 [17.6, 28.9], 65.1 [52.9, 82.7], 69.9 [55.4, 90.1], 77.8 [62.0, 99.8]. The Proth controls: 25.9, 27.2, 88.2, 133.7 (mitm). A/A (rho-neg-aa / rho-neg) = 1.0000 on all four sessions. Files: tables/s*.compare.*.txt.

Order across the band. `rho-neg < ic-mitm` and `rho-neg < ic-rootfind` hold at every `w` in {0, 0.1, 1} on all eight curves; the only unresolved pairs are `rho-neg` against its own A/A copy and `ic-mitm` against `ic-rootfind` on the 28-bit Solinas curve (`tables/s*.band.txt`). The F4 arm's order against rho holds at every `w` (its `S` is `w`-independent to 4 digits: `U` = 213,652 unpriced operations per system is 1.1e8 GAE at `w = 1` over 522 relations, against 2.4e12 GAE priced).



ecbench sessions (measured) and the derived F4 lower bound (PROTOCOL 4.4). Solid: Solinas, dashed: Proth.

3.4 Solinas against Proth at the same tower height

Predicted (P6): `S_IC/S_rho` on a Solinas curve within 20% of the Proth control at each size. **Failed as stated**: the ratios differ by -29%, +126%, -24% and -43% at 20, 24, 26, 28 bits (18.4 vs 25.9, 61.4 vs 27.2, 67.0 vs 88.2, 75.9 vs 133.7). The differences are not the prime's shape:

- `t` is chosen by bit size, so a Solinas and a Proth curve of the same size have the same factor-base height, while `r` differs (cofactors 1-4: `log2 r` 19.98 vs 18.85, 22.00 vs 23.38, 25.00 vs 23.28, 28.00 vs 26.20). `S` divides by `sqrt r`.
- **Absolute cost at equal `t`** (derived from the sealed records, 95% bootstrap over workloads, `tables/gae_solinas_vs_proth.md`): Solinas / Proth GAE for `ic-mitm` is 1.053 [1.034, 1.071], 1.324 [1.297, 1.357], 1.206 [1.196, 1.216], 1.017 [1.014, 1.020] at 20, 24, 26, 28 bits, and for `ic-rootfind` 1.034, 1.300, 1.197, 1.016 (same intervals' order). **The Solinas curves cost more in absolute GAE at every size**, 2% to 32%, not less, in both IC arms. Rho moves with `sqrt r` (Solinas/Proth rho GAE 1.48, 0.59, 1.59, 1.79).
- So where the Solinas `S` ratio is lower, rho got more expensive, the index calculus did not get cheaper. By AGENTS.md §3 that is **accounting** (the numbers changed, the algorithm did not) and it is not claimed as an improvement. No row met the protocol's improvement condition (§5 of the protocol) once `r` is accounted for.
- Growth with size (`d log2(S_ic/S_rho) / d log2 r`, 4 sizes, least squares, 95%): Solinas `ic-mitm` 0.217 ± 0.225, `ic-rootfind` 0.192 ± 0.202; Proth `ic-mitm` 0.304 ± 0.350, `ic-rootfind` 0.268 ± 0.347; prediction 0.17, no fit excludes it or zero (`tables/growth_fits.txt`). With `t` tied to `bits` and `r` varying by cofactor, this fit is poorly conditioned; it is reported, not relied on.

3.5 The F4 arm (derived; PROTOCOL §4.4, formula frozen before the runs)

`GAE_F4 = R * (madds / M + w * U)`, `R = columns + 10`, `S = GAE_F4 / sqrt r`.

input	value	label
curve	20-bit Solinas, `r = 1,030,723`, `t = 10` (`N = 20`)	
`columns`	512, the on-curve abscissae of `ic-mitm-tf4` (`prime-kummer:t=10`)	measured (factor-base record)
`madds`	4.819e11 per system, mean of 2 random targets (R4)	measured
`ns / madd`	1.05, `ms / madds` of R4 under load	measured
`ns / GAE`	101-139 across five invocations of the report tool; the table uses the stored one, 108.0 (microbenchmark of the harness's counted `PrimeCurve` addition, median of 5 runs of 10^6 additions; `analysis/ns_per_gae_1032193.txt`)	measured
`M = ns/GAE / ns/madd`	97-133 madds per GAE (the table uses the stored one: 103.2)	derived
`U`	213,652: F4 pairs + reducer rows + max rows per system	measured, priced at `w`
`S_F4`	2.40e9 at every `w`; ratio to rho 1.9e9 / 1.8e9 / 1.0e9	derived, **lower bound**
generous bound	`M = 4` (inversion costed as one multiplication): `S_F4 = 6.2e10`	derived

Lower bound because it counts one F4 solve per relation, 522 relations (`R`); it omits the failed targets, the factor-base build and the linear algebra. A second priced row needs `N = 24` (R18 failed), so there is **no scaling fit** for the F4 arm. Per the protocol the 26- and 28-bit curves, which need `N = 26, 28`, get no F4 row and none is extrapolated. `tables/f4_priced.md`.

Prediction P7 (F4 arm at least 10^5 times rho at `N = 20`): **held** (1.9 × 10^9).

4. Failures and deviations (all preserved)

1. **R18 was OOM-killed (measured, kernel log).** SL24, `m = 2`, `N = 24`, one target, 2 threads, `--max-nnz 1.8e9`, `ulimit -v` 9 GB. Started 04:19 UTC; killed about 06:45 UTC by the memory-cgroup OOM killer, `anon-rss 6,007,580 kB`, the killer having been invoked by another agent's process in the shared container cgroup. The pilot writes a system's row when the system ends, so the run left no row and no step trace: no `D` and no lower bound. Per protocol §2.2 it is recorded as out of budget (memory) and is **not re-run**. The empty files `runs/R18-SL24-m2-N24.jsonl` and `.log` are kept.
2. **R18b (Amendment 2, committed before it ran)** re-ran the same system with a 600M-entry cap and a trace. It stopped by design at step 26 and gave `D >= 6` only (§2).
3. **`progress.txt` exit statuses are void for `run_f4.sh`.** The `echo` printed the `$?` of its own `date` substitution, so every `end ... exit 0` line from that script carries no information (R18 shows `exit 0` though the kernel killed it). Completion is read from each run's summary line in `runs/*.log` (the only runs without one: R18). The script is fixed in this tree; the version that ran had the bug.
4. **Parallel runs.** The main script was stopped by hand after R16 started (the child kept running), and R16n, R17 and R19 were started in parallel with the same flags through `run_parallel.sh` to use idle cores; `progress.txt` has a NOTE (the script was tidied of a redundant `case` after the runs, same flags). The cells and flags are the protocol's; only the order and concurrency changed.

5. **Load.** The host's load average was 4-10 throughout, shared with other agents, so wall times (`ms`) are descriptive only; operation counts do not depend on it.
6. **ecbench ran at L0**, `--cpus none --allow-busy`, so no wall time is admitted; operation counts, audits and replays are unaffected. The spec says `isolation_required: L2`; the harness ran and graded every run L0.
7. **Scratch smoke runs.** Before the first run of each tool the author ran short smoke tests of the pilot at SL26 and P-224 (`N <= 6`, 14 rows) in a scratch directory; they are not results and are not in this directory.
8. **A transcription slip in the report tool**: the recorded `D` at `N = 10` was first coded as 5; the protocol says 4, and three cells were reported as `H_diff` in a trial print before the fix. The committed output is from the fixed tool; no run changed.
9. **Curves not registered** (Amendment 3 in the protocol; §3.2 says why).

5. Predictions against outcomes

	prediction	outcome
P1	R0 gives `D = 5`, `N = 4-10`	**held**
P2	H_same on every valued cell; `D` rises to 6 at `N = 20` at every prime	**held** (89 of 89)
P3	`D(N = 24)` is 6 or 7	**no value** (R18 killed; R18b `D >= 6`)
P4	tower and null agree at every cell	**held** (65 of 65)
P5	multiply-adds equal across backends at one thread	**held** (to 3 × 10^-6)
P6	Solinas S ratio within 20% of Proth	**failed as stated**, explained by `r` (§3.4)
P7	F4 arm at least 10^5 x rho	**held**

6. What this shows, what it does not, and the next step

Shows (measured). The F4 solving degree and the whole F4 computation do not see a P-224-like ratio of 2-adic to odd part: equal `D`, width, steps and multiply-adds at eight Solinas primes and at P-224's. The tower family still reads inconclusive at `N <= 22`. Priced end to end the PKM F4 method costs 10^9 times rho at the one size that matches a curve, and the table oracles on the same base cost 18-134 times rho and the gap grows slowly with size.

Does not show. Anything at `N` near P-224's size, `m >= 4` at `N` above 12, whether `D` rises past 6 at `m = 2`, or a priced F4 row at a second size. Nothing here attacks P-224; a bounded null is not nonexistence (AGENTS.md).

Next improvement to try (each is a different experiment, not done here):

1. `m = 2, N = 24` and `m = 3, N = 21` on a host with 64 GB or more (`B'` is 4 bytes per entry; R18b's step 26 alone is 3.1 GB, and the run died at 6.0 GB resident), which would answer Q2 and give the F4 arm a second priced size.
2. An `ic.pipeline` oracle that calls `f4_tower_generic`, so the F4 arm is measured in the harness (`ecbench-extend`) instead of derived.
3. Curves matched on `r`, not on bits, for any Solinas-against-Proth `S` comparison, since §3.4 shows the bit-matched one is confounded.

7. Environment and reproduce

Claude Code cloud container, 4 vCPU Xeon @ 2.10 GHz, 15 GiB RAM shared with other agents, rustc 1.97.0, binary `pkm_tower_pilot` sha256 `aab76a30ab352de3…` (runs/BINARY.sha256), built from commit `dd0486c6d` (the pilot source is unchanged from `main`). Conductor `check` reported the scope clear.

```
        export CARGO_TARGET_DIR=/path/to/target
        cargo build --release --example pkm_tower_pilot --example pkm_tower_check \
          --example prime_tower_curves --example pkm_p224_shape_report --bin ecbench
        research/pkm_p224_kummer_shape_20261022/make_specs.sh          # specs from
curves.jsonl
        BIN=$CARGO_TARGET_DIR/release/examples/pkm_tower_pilot
research/pkm_p224_kummer_shape_20261022/run_f4.sh main
        BIN=... research/pkm_p224_kummer_shape_20261022/run_f4.sh frontier   # R18 (needs
~10 GB)
        BIN=... research/pkm_p224_kummer_shape_20261022/run_r18b.sh
        ECBENCH=$CARGO_TARGET_DIR/release/ecbench
research/pkm_p224_kummer_shape_20261022/run_sessions.sh
        pkm_p224_shape_report f4 runs .  &&  pkm_p224_shape_report price runs . curves.jsonl
sessions/s20 sessions/s24 sessions/s26 sessions/s28
        research/pkm_p224_kummer_shape_20261022/render.sh              # PNGs and report.pdf
```

Curves: `prime_tower_curves --solinas 20:14,24:14,26:12,28:16 --seed 20261022 --max-cofactor 4` (curves.jsonl, curves.log). Specs: specs/. Sealed sessions: sessions/; audits (audits/): s20 and s24 320 records, 240 replays; s26 and s28 256 records, 192 replays; every record verified, every replay reproduced, 0 problems. Run rows and logs: runs/. Report PDF: report.pdf.